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Division: **Neurons**

App ID: **2509399**

1. Display the average salary for each department.

```
SQL> SELECT dept_id_09399, AVG(monthly_salary_09399) AS avg_salary  
2 FROM staff_09399  
3 GROUP BY dept_id_09399;
```

DEPT_ID_09399	AVG_SALARY
1	40000
11	8555.5
6	7800
17	6000
7	5500
3	2800
12	5333.33333

7 rows selected.

SQL>

2. Find the maximum salary, minimum salary and total salary given for every job_title in every department.

```
SQL> SELECT dept_id_09399,
2         job_title_09399,
3         MAX(monthly_salary_09399) AS max_salary,
4         MIN(monthly_salary_09399) AS min_salary,
5         SUM(monthly_salary_09399) AS total_salary
6 FROM staff_09399
7 GROUP BY dept_id_09399, job_title_09399;
```

DEPT_ID_09399	JOB_TITLE_09399	MAX_SALARY	MIN_SALARY	TOTAL_SALARY
17	SALESMAN	1000	1000	1000
12	DEVELOPER	11000	11000	11000
1	CLERK	70000	70000	70000
17	MANAGER	11000	11000	11000
12	MANAGER	2000	2000	2000
7	ANALYST	8000	8000	8000
12	SALESPERSON	3000	3000	3000
11	PRESIDENT	11111	11111	11111
1	DEVELOPER	10000	10000	10000
3	DEVELOPER	4500	4500	4500
6	CLERK	3600	3600	3600

DEPT_ID_09399	JOB_TITLE_09399	MAX_SALARY	MIN_SALARY	TOTAL_SALARY
6	DEVELOPER	12000	12000	12000
7	CLERK	3000	3000	3000
11	MANAGER	6000	6000	6000
3	SALESPERSON	1100	1100	1100

15 rows selected.

3. Find departments where the total commission is more than 1000.

```
15 rows selected.

SQL> SELECT dept_id_09399,
2         SUM(commission_amt_09399) AS total_commission
3 FROM staff_09399
4 GROUP BY dept_id_09399
5 HAVING SUM(commission_amt_09399) > 1000;
```

DEPT_ID_09399	TOTAL_COMMISSION
11	6111
6	6000
17	11000
3	1200

4. Display departments having an average salary less than 4000.

```
SQL> SELECT dept_id_09399,
2          AVG(monthly_salary_09399) AS avg_salary
3 FROM staff_09399
4 GROUP BY dept_id_09399
5 HAVING AVG(monthly_salary_09399) < 4000;

DEPT_ID_09399  AVG_SALARY
-----
3              2800

SQL>
```

5. Display employees with commission, ordered by commission amount (highest first).

```
SQL> set pagesize 100
SQL> set linesize 200
SQL> SELECT *
2 FROM staff_09399
3 WHERE commission_amt_09399 IS NOT NULL
4 ORDER BY commission_amt_09399 DESC;

STAFF_ID_09399  FULL_NAME_09399                                JOB_TITLE_09399
MANAGER_ID_09399  DATE_HIRE  MONTHLY_SALARY_09399  COMMISSION_AMT_09399  DEPT_ID_09399
-----
13 MAQ          11 22-MAR-11          11000          11000          17
14 GAUSIYA      7 05-FEB-11          12000          6000           6
6 NASHRA        1 18-FEB-11           6000          5000          11
5 LION          12 30-SEP-11          4500          1200           3
11 ELEVEN       11-NOV-11            11111          1111          11
8 KHADIJAH      12 10-OCT-11          10000          1000           1
7 MADINAH       7 28-JAN-11          11000           500          12
4 FOX           6 20-MAY-11           3000          111           7

8 rows selected.

SQL>
```

6. Show departments in alphabetical order of department name.

```
SQL> SELECT DISTINCT dept_id_09399, dept_name_09399
  2  FROM department_09399
  3  ORDER BY dept_name_09399 ASC;

DEPT_ID_09399 DEPT_NAME_
-----
          11 Finance
           7 HR
          12 HR
           1 Marketing
          17 Marketing
           3 Sales
           6 Sales

7 rows selected.

SQL>
```

7. Find the average salary per department and display only those departments having average salary > 4000, ordered by average salary descending.

```
SQL> SELECT dept_id_09399,
  2      AVG(monthly_salary_09399) AS avg_salary
  3  FROM staff_09399
  4  GROUP BY dept_id_09399
  5  HAVING AVG(monthly_salary_09399) > 4000
  6  ORDER BY avg_salary DESC;

DEPT_ID_09399 AVG_SALARY
-----
           1      40000
          11    8555.5
           6      7800
          17      6000
           7      5500
          12 5333.33333

6 rows selected.

SQL>
```

8. List job_titles and number of employees in each, but only display job_titles that have at least 2 employees, ordered alphabetically.

```
SQL> SELECT job_title_09399,  
2         COUNT(*) AS num_employees  
3 FROM staff_09399  
4 GROUP BY job_title_09399  
5 HAVING COUNT(*) >= 2  
6 ORDER BY job_title_09399 ASC;
```

JOB_TITLE_09399	NUM_EMPLOYEES
CLERK	3
DEVELOPER	4
MANAGER	3
SALESPERSON	2

```
SQL>
```

9. Display each department's total salary bill (sum of salaries) and order by highest to lowest.

```
SQL> SELECT dept_id_09399,  
2         SUM(monthly_salary_09399) AS total_salary  
3 FROM staff_09399  
4 GROUP BY dept_id_09399  
5 ORDER BY total_salary DESC;
```

DEPT_ID_09399	TOTAL_SALARY
1	80000
11	17111
12	16000
6	15600
17	12000
7	11000
3	5600

```
7 rows selected.  
SQL> _
```

10. Show the minimum salary per department and display only those where min salary > 2000, sorted in ascending order of min salary.

```
SQL> SELECT dept_id_09399,  
2      MIN(monthly_salary_09399) AS min_salary  
3 FROM staff_09399  
4 GROUP BY dept_id_09399  
5 HAVING MIN(monthly_salary_09399) > 2000  
6 ORDER BY min_salary ASC;
```

DEPT_ID_09399	MIN_SALARY
7	3000
6	3600
11	6000
1	10000

```
SQL>
```

11. Find the average salary of each department, but only include employees hired after 01-JAN-2011, and display only those departments where the average salary is greater than 4000.

```
SQL> SELECT dept_id_09399,  
2      AVG(monthly_salary_09399) AS avg_salary  
3 FROM staff_09399  
4 WHERE date_hired_09399 > DATE '2011-01-01' GROUP BY dept_id_09399  
5 HAVING AVG(monthly_salary_09399) > 4000;
```

DEPT_ID_09399	AVG_SALARY
1	40000
11	8555.5
6	7800
17	6000
7	5500
12	5333.33333

```
6 rows selected.
```

12. Display the total commission per job_title, but only consider employees whose salary > 2000, and show only those job_titles where the total commission is greater than 500.

```
SQL> SELECT job_title_09399,  
2          SUM(commission_amt_09399) AS total_commission  
3 FROM staff_09399  
4 WHERE monthly_salary_09399 > 2000  
5 GROUP BY job_title_09399  
6 HAVING SUM(commission_amt_09399) > 500;
```

JOB_TITLE_09399	TOTAL_COMMISSION
PRESIDENT	1111
MANAGER	16000
DEVELOPER	8700

```
SQL>
```